Autonomous Closed-Loop Discovery of Additively Manufactured Alloys for Extreme Aerospace Environments

AFOSR Young Investigator Program — Pre-solicitation White Paper

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AFOSR portfolio: Aerospace Materials for Extreme Environments (Dr. Ali Sayir, Program Officer)
Requested: ~$450K over 36 months (single PI, no subawards)

Problem and Air Force relevance

Air Force propulsion and hypersonic platforms are limited by the materials at their hottest surfaces—turbine hot-section superalloys and the refractory alloys of hypersonic leading edges. Designing such alloys is slow and strategically fragile: the best nickel superalloys depend on supply-constrained, largely foreign-sourced cobalt, rhenium, and hafnium, and the refractory complex-concentrated-alloy (RCCA) space for 1600 °Cplus service is vast, sparsely characterized, and largely unexplored by hand. There is a tight bottleneck in *decision-making*: which of the many compositions and process windows to evaluate next. This effort targets that bottleneck directly.

Objective

Establish an autonomous, closed-loop discovery platform that selects, fabricates, and characterizes alloy compositions and laser powder-bed-fusion (LPBF) process conditions to maximize high-temperature capability per experiment while explicitly penalizing strategic-element content. The two primary deliverables are (1) a transferable and extensible platform for autonomous alloy discovery and (2) the implementation and validation of this platform within an elemental palette consisting of alloys relevant to the Air Force.

Technical approach

The platform couples simulation, machine learning, and automated experiment in one loop:

- Multi-fidelity Bayesian optimization over a function network (BOFN). The powder-to-property pipeline (CALPHAD → atomization → LPBF → microstructure → property) is modeled as a directed acyclic graph of surrogate models, so the algorithm strategically chooses when to query cheap nodes (thermodynamics simulations and lower-cost proxy experiments) vs. expensive ones (full tensile test) based on the information gained relative to the cost to gain that information. This cost-aware, partial-evaluation formulation is the core methodological novelty.
- CALPHAD-anchored screening (pycalphad / Thermo-Calc, refractory and Ni databases) supplies highthroughput, physics-grounded priors over phase stability and processability before any powder is made.
- Automated fabrication and characterization on plasma-based ultrasonic atomization, in-house BYU LPBF, SEM/EDS/EBSD, and elevated temperature mechanical testing

Demonstration system. *Primary:* refractory complex-concentrated alloys (Mo–Nb–Ta–W–Hf–Zr–Ti) for hypersonic leading edges. *Secondary:* strategic-element-lean Ni-superalloy variants for turbine hot sections, holding creep/oxidation resistance while cutting Co/Re/Hf burden (shared methodology for either).

Preliminary results and PI qualifications

I have a working multi-fidelity BOFN implementation, a CALPHAD pipeline running on BYU high-performance computing, and a powder-to-part demonstration on a baseline aluminum system (atomized feedstock, LPBF builds, SEM/EDS, and tensile data in hand). Since my hire six months ago, we collected this data to help vet and de-risk the feasibility of the workflow.